Arundhati Sonawane

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.70
DPS, Doha-Qatar	Class XII (CBSE)	2020	86.40%
DPS, Doha-Qatar	Class X (CBSE)	2018	92.20%

PROJECTS

Operational mode classification and Fuel Efficiency Forecasting | Introduction to ML and its applications | IIT Hyderabad |

- Developed an RNN-based framework to classify automotive engine operating modes--*Idle, Cruising, and Heavy Load*--using temperature, RPM, torque, power, and 3-axis vibration data.
- Implemented sequence generation, dropout and Adam optimization, achieving 0.29 validation classification accuracy and analyzing performance using precision, recall, F1-score, and confusion matrix.
- Built a Random Forest regression model for fuel-efficiency forecasting (RMSE = 1.65, R^2 = 0.86) and integrated rule-based recommendations for intelligent driving and predictive-maintenance insights.

ABAQUS-MATLAB plugin to perform Multi-material Topology Optimization | M.tech thesis | IIT Hyderabad |

- Development of a computational design and structural optimization approach for lightweight automotive components, with a focus on improving structural performance while reducing material usage.
- The project explores the integration of finite element analysis and numerical optimization to efficiently develop optimized designs for components with complex and practical geometries.

Composite Material Characterization (Nov 2023-May 2024) | B.E Project | MIT Chennai | IIT Madras |

- Fabricated and characterized hybrid Ramie-PVA composite laminate reinforced with Carbon and Basalt fibers.
- Performed tensile testing using UTM and utilized Digital Image Correlation (DIC) to analyze deformation behavior.
- Performed fractographic analysis to identify failure mechanisms and systematically compared the mechanical properties of the reinforcement.

Structural Topology Optimization of Aircraft Wing Leading Edge | Review presentation | Topology Optimization | IITH |

- Understanding of how optimal internal structure of a leading-edge slat using topology optimization, while satisfying material cost limitations is identified.
- Comparing the original metallic slat structure, the TO slat with sandwich shell reduces structural weight by 28.9%.

FEA and material selection for automotive wheel rim | Introduction to LW | IIT Hyderabd|

- Performed finite element analysis under radial loading and tyre inflation pressure to evaluate equivalent stress and total deformation
- Evaluated lightweighting potential using weight and Factor of Safety, achieving a reduction from 21.03 kg (Al) to 11.65 kg (Carbon Fibre Reinforced Polymer) with FoS of 1.27 and 1.78, respectively.

OBJECTIVE

Life is too short to solve problems with conventional methods, boxing it out to venture into the world of lightweight.

EXPERIENCE

University of Siegen | 6-month exchange program| Nov'26- April'27|

DYNAMIC MECHANICAL TESTING | Summer Internship Pragram | IIT Madras| 15.06.2023 to 26.07.2023|

- Developed a dynamic mechanical testing setup for characterizing viscoelastic rubber specimens under dynamic loading.
- Integrated NI DAQ, strain-gauge load cell, piezoelectric actuator, displacement sensor, and LabVIEW for real-time force/displacement data acquisition.
- Calibrated force and displacement sensors and generated stress-strain responses for rubber specimens at different test conditions.

DEF-SPACE Summer Internship | Bharat Space Education Research Centre | (REMOTE)| June 19 to August 9, 2026|

- Analysed emerging aerospace and defence systems, specialising in domains such as advanced rocketry, defence drone technology, aircraft design, and robotics frameworks.
- Developed project solution utilizing AI/ML.

Rocket Design Intern | Pegasus Aerospace| (REMOTE)| 01/08/2022 to 01/09/2022 |

- Gained an understanding of conceptual design and analysis of rocket systems.

RELEVANT COURSES AND LANGUAGE

Courses: Fundamentals of Light Weight Alloys, Introduction to Lightweighting, Topology Optimization, Overview of Composites, Soft Robotics, Advanced Finite Element Method, Introduction to Machine Learning and its Application, Additive Manufacturing, Lightweight Design topics lab

Languages: English, Hindi, Marathi, Tamil

SKILLS

MATLAB, SOLIDWORKS, ABAQUS, ANSYS, Materialise MAGIC, Python

POSITIONS OF RESPONSIBILITY AND ACTIVITIES

- Placement Representative| 2023-2024 | MIT, Chennai |

- Technical Games Head, 2024| FLIGHT'24 | MIT, Chennai |

- Joint Treasurer, 2023: The Box Office Club | MIT, Chennai |

- Inter-IIT, 2025| Basketball Player | IIT Hyderabad |

- Second'2022 | Basketball Inter-Zonals | MIT Chennai |